Well-Architected Frameworks

six lenses on the same design

Watch the 4-min video

In one line: AWS, Azure and Google Cloud each publish a well-architected framework: pillars (reliability, security, cost, operations, performance, and on AWS sustainability), design principles and review questions, used as **lenses** on one design to surface risks and trade-offs before customers find them.

1 Two weeks before launch

Your architect asks three questions. *What happens if this region goes down?* Nobody knows. *Who can delete the production database?* Eleven people. *What will this cost at ten times the traffic?* Silence. None of these are exotic; they are simply questions nobody was responsible for. A well-architected review is a structured way to ask them all, every time, and to record the answers.

2 The frameworks and their pillars

Provider	Pillars	Tools
AWS Well-Architected	Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability (added 2021)	Well-Architected Tool (marks high and medium risk issues); lenses such as SaaS and Serverless
Azure Well-Architected	Reliability, Security, Cost Optimization, Operational Excellence, Performance Efficiency	Well-Architected Review assessment; Azure Advisor recommendations by pillar
Google Cloud (Architecture / Well-Architected Framework)	the same core areas, with security combined with privacy and compliance	framework guides and design checklists

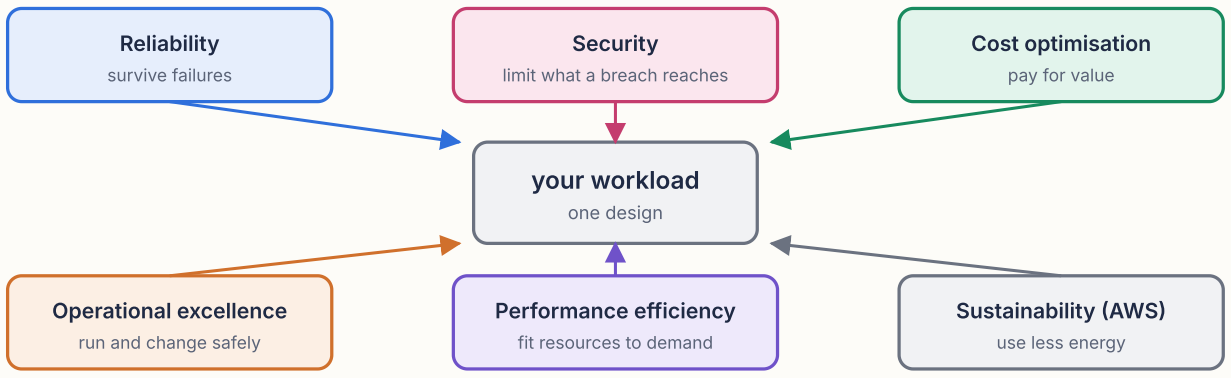


Figure 1. The pillars are not separate checklists to tick once; each is a different way of looking at the same system.

3 Reliability and security

Reliability is about recovering from failure and meeting demand. Set recovery targets per workload: **RTO** (how long you may be down) and **RPO** (how much data you may lose). Then design to them: redundancy

across availability zones (lesson 52), health checks and automatic failover, timeouts and retries with backoff, and backups you have **actually restored**. The key review question: what is our RTO, and have we proven we can meet it with a drill?

Security covers identity, data and network. Least privilege (no standing owner rights, short-lived credentials, managed identities instead of keys), encryption in transit (TLS) and at rest, network isolation (lesson 53), secrets in a vault, and detection: audit logs and alerts on suspicious activity. **Assume breach**: for each component, ask what an attacker who controls it could reach next, and shrink that.

4 Cost optimisation and operational excellence

Cost optimisation means paying only for what delivers value: right-sizing (many instances run at low average CPU), autoscaling, reservations or savings plans for steady load, spot capacity for interruptible work, storage tiering, and deleting what nobody uses. Measure **unit cost**, such as cost per tenant or per 1,000 requests, so growth and waste can be told apart. Tag every resource with an owner, environment and cost centre, and set budget alerts. More in lessons 64 and 65.

Operational excellence is how you run and change the system: infrastructure as code (lesson 61), CI/CD with safe deployment (canaries, feature flags, fast rollback), monitoring and alerting on user-facing SLOs, runbooks, and blameless post-incident reviews. Prefer small, frequent, reversible changes; the DORA metrics (deployment frequency, lead time for changes, change failure rate, time to restore) are a common way to measure it.

5 Performance efficiency and sustainability

Performance efficiency is using resources well as demand changes: choosing the right compute (CPU, memory or GPU optimised; ARM-based instances often give better price-performance), storage and database types, caching, CDNs, and **load testing** before launch against the expected peak. Measure before scaling: slow requests usually come from the data layer (missing indexes, N+1 queries), not from too few servers.

Sustainability, AWS's sixth pillar, asks you to minimise the energy and hardware a workload consumes: raise utilisation, remove idle and over-provisioned capacity, scale to zero where possible, keep only the data you need, and prefer efficient hardware and regions with lower-carbon electricity. It mostly aligns with cost: the greenest server is often the one you can switch off (lesson 73).

6 The pillars pull against each other

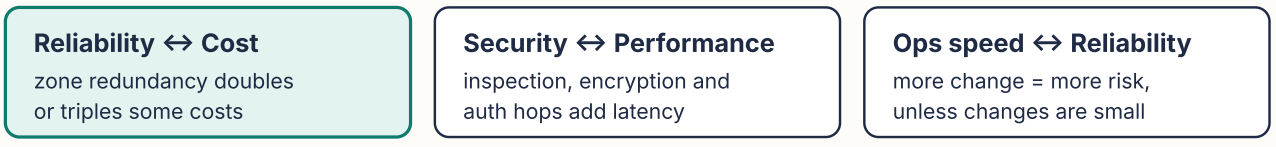


Figure 2. Typical tensions. A review doesn't remove them; it makes the choice explicit and documented.

More redundancy raises reliability and cost. Deeper inspection raises security and latency. Faster change raises delivery speed and, without guard-rails, risk. The value of a review is turning accidental trade-offs into deliberate ones: "single-zone dev environment, accepted, because it saves 40% and has no SLA" is a decision; nobody noticing is not.



Figure 3. The review loop. Findings become a tracked backlog, not a slide.

WORKED EXAMPLE: A REVIEW BEFORE LAUNCH

The team answers the Azure Well-Architected Review questions for each pillar, with evidence (screenshots, configs, drill results). Findings: the Azure SQL database is not zone-redundant; eleven people hold Owner on the production subscription; no restore has ever been tested; there are no budget alerts; there's no load test above 2× current traffic. They fix the high-risk items before launch: enable zone redundancy, replace standing Owner rights with least-privilege roles and just-in-time elevation, and run a restore drill that measures a real RTO of 35 minutes. Budget alerts and a 10× load test go on the backlog with owners and dates. Six months later the re-review list is much shorter.

COMMON MISTAKES

- Treating the framework as a one-off checklist instead of a recurring review.
- Answering "yes, we have backups" without having restored one.
- Reviewing only reliability and security, never cost and operations.
- Recording findings without owners, priorities and dates.

7 Key terms

Pillar: one lens: reliability, security, cost, ops, performance, sustainability

Least privilege: only the access a role needs, only when needed

High-risk issue: finding to fix before launch

RTO / RPO: max downtime / max data loss

Unit cost: cost per tenant, request or transaction

DORA metrics: four measures of delivery performance

8 Interview questions

Q1. Name the well-architected pillars. BASIC

Reliability, security, cost optimisation, operational excellence, performance efficiency, and on AWS sustainability.

Q2. One practice for each? BASIC

Zone redundancy with tested restores; least privilege and encryption; right-sizing and autoscaling; infrastructure as code with safe deployments; caching and load testing; higher utilisation with no idle resources.

Q3. You have one week before launch. Which review findings do you fix first? SCENARIO

The high-risk ones that cause data loss, outages or breaches: untested restores, single-zone databases, excessive admin rights, public exposure; schedule the rest with owners.

Q4. How do you show a reliability design actually works? SCENARIO

Define RTO and RPO, then prove them with restore drills, failover tests and chaos experiments, and record the measured times.

Q5. Give an example where two pillars conflict and how you'd decide. TRICKY

Multi-region active-active improves reliability but roughly doubles cost and adds consistency complexity; decide from the business's RTO, RPO and budget, and document it.

Q6. Is a well-architected review a compliance audit? TRICKY

No: it's a self-assessment against best practices to find risks and trade-offs; compliance audits (SOC 2, ISO 27001) check controls against a standard.

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